

TrajRAG: Textual State-Driven Retrieval-Augmented Iterative Reconstruction of Spoofed Aircraft Trajectories

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Abstract—Global Navigation Satellite System (GNSS) interference can corrupt aircraft positions broadcast through Automatic Dependent Surveillance–Broadcast (ADS-B), creating periods in which horizontal navigation data cannot be trusted. While prior work has largely focused on detecting such events, trajectory recovery after detection remains less studied. We introduce TrajRAG, a retrieval-augmented framework for post-detection trajectory reconstruction. Rather than extrapolating untrusted observations or retraining a task-specific predictor, TrajRAG grounds each reconstruction step in historically observed, phase-consistent flight behavior. ADS-B and weather sequences are converted into structured textual states and stored in phase-aware vector indices; retrieved segments condition an LLM next-state proposal, which is then constrained by deterministic kinematic validation. We evaluate TrajRAG on 1,544 real commercial flights across the Paris–Rome and Barcelona–Munich corridors under simulated GNSS blackout intervals, together with four MLAT-supported real-world interference cases. Across flight phases, TrajRAG reduces mean and worst-case reconstruction error relative to classical, neural, modern time-series, and retrieval-only baselines, while remaining practical under real-time constraints and mixed-route memory expansion. These results support retrieval-grounded reconstruction as a practical post-detection mechanism for maintaining trajectory continuity when GNSS-derived horizontal positions cannot be trusted.

Index Terms—GNSS Spoofing, ADS-B, Trajectory Reconstruction, Retrieval-Augmented Generation, Aviation Security

I. INTRODUCTION

Global Navigation Satellite System (GNSS) spoofing poses a growing risk to civil aviation by corrupting aircraft navigation states derived from satellite signals [1]. When spoofing occurs, incorrect position and velocity estimates propagate through Automatic Dependent Surveillance–Broadcast (ADS-B) messages, degrading situational awareness for air traffic controllers and downstream decision-support systems [2], [3].

While substantial effort has focused on the detection of GNSS anomalies, a critical operational challenge remains: the reconstruction of aircraft trajectories during intervals in which navigation data cannot be trusted. When interference is detected, corrupted position, navigation, timing (PNT) measurements must be discarded, resulting in a temporary loss of reliable horizontal positioning.

Current solutions mainly rely on trajectory prediction models used as extrapolation mechanisms. Once interference is detected, however, corrupted horizontal observations can no

longer be safely assimilated, forcing prediction to proceed open-loop from the last verified state. Error can therefore accumulate over longer gaps, while a single learned model must cover phase- and route-dependent dynamics and may require retraining as operating conditions change [4], [5].

This motivates a different formulation: trajectory recovery under navigation blackout as retrieval-grounded sequential inference rather than pure extrapolation. Each prediction step is anchored to historically observed, phase-consistent flight behavior, allowing the reconstruction process to adapt through memory retrieval rather than parameter updates.

We introduce **TrajRAG**, a retrieval-augmented framework for aircraft trajectory reconstruction during GNSS-induced navigation blackout. Instead of extrapolating corrupted observations or retraining task-specific predictors, TrajRAG reconstructs motion by conditioning Large-Language-Model (LLM) predictions on semantically similar historical trajectory segments retrieved from a phase-aware vector index. The proposed framework operates at inference time, ensuring data efficiency and robustness to distribution shift. The framework is designed for operational deployment in aviation monitoring pipelines, where GNSS interference detection triggers trajectory reconstruction.

A central challenge in enabling retrieval on ADS-B streams is representation. TrajRAG converts trajectory segments into structured textual state representations summarizing spatial evolution, kinematic trends, flight phase, and aligned meteorological context. These representations facilitate semantic embedding, contextual similarity search, and interpretable reasoning [6].

TrajRAG segments trajectories into takeoff, cruise, and landing phases to account for heterogeneous aircraft dynamics. The segmentation is done using barometric altitude trends, which remain unaffected by GNSS spoofing [7]. Separate phase-specific vector indices prevent mixing incompatible motion patterns. During reconstruction, the framework initializes from the last verified state and iteratively alternates between retrieval and state generation, with deterministic kinematic validation constraining physically infeasible motion and mitigating LLM hallucinations. The contribution of TrajRAG is not RAG alone, but a reconstruction pipeline organized around the post-detection trust boundary: GNSS-derived horizontal

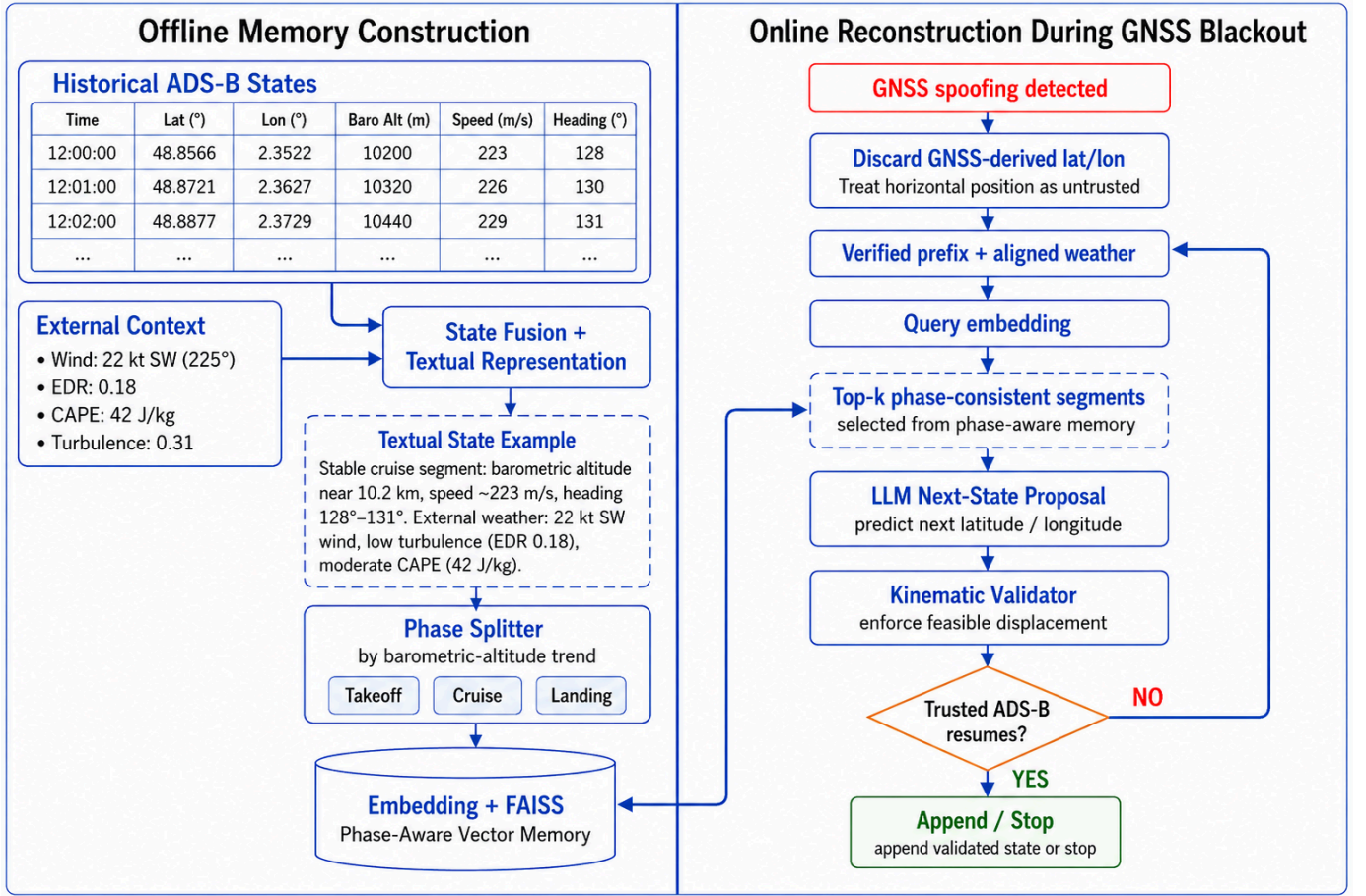


Fig. 1. TrajRAG pipeline. Historical ADS-B and weather records are converted into phase-specific textual state representations and indexed offline. During a post-detection GNSS blackout, the verified prefix is embedded as a query, top-k phase-consistent segments are retrieved, and an LLM proposes the next state, which is accepted only after kinematic validation.

observations are discarded rather than denoised, retrieval is restricted to phase-consistent historical behavior, and each LLM-generated position is treated only as a proposal subject to deterministic kinematic validation. This combines non-parametric memory with aviation-specific constraints so that adaptation occurs at inference time without retraining. We evaluate TrajRAG on 1,544 real commercial flights across two European corridors under simulated GNSS corruption conditions. Compared to classical filtering, statistical forecasting, recurrent sequence models, modern time-series forecasting baselines including DLinear, PatchTST, and iTransformer, and retrieval-based alternatives, TrajRAG consistently achieves lower mean reconstruction error and substantially reduces worst-case deviations, while remaining robust under mixed-route memory expansion and demonstrating superior data efficiency.

II. RELATED WORK

A. ADS-B and GNSS Interference

ADS-B is a widely deployed cooperative surveillance technology in which aircraft broadcast position, velocity, and

barometric altitude to ground stations and nearby aircraft [8]–[10]. Because ADS-B horizontal position and velocity are derived from onboard GNSS receivers, GNSS interference propagates directly into ADS-B streams [11], [12]. Recent reports document increasing GNSS anomalies affecting civil aviation, highlighting the need not only for spoofing detection but for reliable trajectory recovery once positioning becomes untrusted [1], [13]–[15].

B. Trajectory Reconstruction and Prediction Baselines

Aircraft trajectory prediction has been approached with model-based, statistical, and learning-based methods; recent surveys group deep learning approaches into recurrent, attention-based, generative, graph-based, and hybrid families [16]. Classical trackers such as Kalman and IMM filters perform well for short-horizon extrapolation under bounded-noise assumptions [17], [18], but under GNSS interference corrupted observations must be discarded rather than smoothed, forcing open-loop operation in which error accumulates over long gaps. ARIMA-style models share this limitation and adapt poorly to phase transitions and weather-dependent dynamics [19], [20], while LSTM/BiLSTM models [21], [22]

provide stronger nonlinear baselines but depend on training-distribution coverage and may drift during long blackouts without trusted feedback.

Modern time-series forecasting contributes strong generic baselines: DLinear, a competitive decomposition-based linear model [23]; PatchTST, which tokenizes series into patches for longer context [24]; and iTransformer, which treats variables rather than time steps as tokens to capture dependencies among correlated trajectory variables (longitude, latitude, altitude, speed, heading) [25]—a design well matched to ADS-B’s coupled multivariate motion and recently adapted, including in multi-agent form, to aircraft trajectory prediction [26]–[28]. We accordingly include DLinear, PatchTST, and iTransformer as baselines. All assume continuous, trusted observations; TrajRAG instead targets the post-detection setting, discarding untrusted horizontal fields and grounding reconstruction in phase-consistent historical segments retrieved at inference time rather than in a global parametric predictor.

C. LLMs and Retrieval-Augmented Reconstruction

LLMs have been explored for flight trajectory prediction by converting ADS-B states into prompts and fine-tuning open-source models for single- and multi-step forecasting [29], showing that language-based representations encode useful spatiotemporal patterns—but still through parameter updates and in standard settings where observations remain available. TrajRAG does not fine-tune the LLM, using it only as a retrieval-conditioned proposal generator and achieving task adaptation through phase-aware retrieval, weather-enriched textual states, and post-generation kinematic validation. More broadly, Retrieval-Augmented Generation grounds parametric generation in non-parametric memory [30] and has been extended to general time-series forecasting [31], but such methods do not encode aviation constraints such as flight phase, barometric-altitude consistency, or feasible displacement. TrajRAG adds these by assigning phase from barometric altitude, retrieving only phase-consistent segments, and validating each state against phase-specific kinematic bounds.

III. METHODOLOGY

A. Problem Formulation

Let an ADS-B trajectory be a time-ordered sequence $X = \{x_0, \dots, x_n\}$, where each state x_i contains latitude, longitude, barometric altitude, ground speed, heading, and timestamp. We assume an external detector flags GNSS interference at index t . The verified prefix $X_{0:t}$ is trusted. After detection, post-detection ADS-B latitude, longitude, ground-speed, and heading values are not supplied to the reconstruction loop; GNSS-derived horizontal observations are treated as untrusted and removed. The LLM generates only the next latitude and longitude. Ground speed and heading are subsequently inferred from the accepted reconstructed motion rather than predicted by the LLM or read from post-detection ADS-B measurements.

TrajRAG receives four inputs: (i) the verified ADS-B prefix $X_{0:t}$, including the last trusted motion state, (ii) barometric

altitude and timestamps that remain available independently of the discarded horizontal GNSS fields, (iii) external meteorological context W , queried using the current trusted or reconstructed position, and (iv) a phase-specific historical memory $\mathcal{M}_p$. The last trusted ground speed is retained for initialization of the kinematic bound. The output is a reconstructed horizontal trajectory $\hat{Y}_{t+1:r} = \{(\hat{\phi}_i, \hat{\lambda}_i)\}_{i=t+1}^r$, where r is the time at which trusted reporting resumes or the flight terminates. This setting differs from denoising and standard forecasting: after detection, corrupted horizontal observations are not corrected but discarded. Reconstruction must therefore proceed from the verified prefix, historical memory, and trusted auxiliary signals without access to ground-truth positions during the blackout interval.

B. Method Overview

TrajRAG formulates reconstruction as a retrieval-grounded sequential inference problem. Instead of extrapolating corrupted observations or training a task-specific predictor or adapting LLM parameters through fine-tuning, the method grounds each reconstruction step in historically observed flight behavior. The framework consists of two stages (Fig. 1):

Offline Learning. Historical trajectories are segmented by flight phase and converted into structured textual state representations. These representations are embedded and indexed in phase-aware vector databases.

Online Inference. When spoofing is detected, the reliable trajectory prefix before spoofing is transformed into a textual representation together with the externally queried weather context aligned to the current trusted or reconstructed position, timestamp, and barometric altitude, and used to retrieve the top- k most similar historical segments from the relevant phase-specific index. The LLM is not fine-tuned; instead, it acts as a retrieval-conditioned proposal generator. Given the verified prefix, retrieved historical references, weather context, and a fixed output schema, it proposes the next plausible aircraft state, specifically the next latitude and longitude. Task adaptation is therefore achieved through retrieval grounding, structured prompting, and post-generation validation rather than parameter updates. Each generated state is then passed through a phase-specific kinematic validation layer, which enforces physically plausible displacement derived from historical flight behavior. Validated states are appended to the reconstructed trajectory, and the process iterates until trusted reporting resumes.

C. Textual State Representation

Raw ADS-B trajectories consist of heterogeneous numerical signals and variable-length sequences. To enable semantic retrieval, we convert trajectory segments into structured textual state descriptions that summarize the spatial evolution, kinematic trends, current flight phase dynamics, and aligned environmental context. Unix timestamps are normalized to time-from-takeoff for improved temporal consistency.

For example, a cruise segment is represented as: “Phase: cruise; time-from-takeoff: 42 min; altitude: stable at 34,000

ft; motion: northeast with stable heading; ground speed: 448 kt; recent trend: no abrupt heading or altitude change; weather: moderate tailwind, low turbulence indicators.” This abstraction preserves numerical trends while exposing operational context useful for retrieval. The textual representation produces an interpretable description of flight behavior that enables semantic retrieval, similarity search across trajectories of different lengths and routes, and effective LLM reasoning. Aircraft dynamics differ during takeoff, cruising, and landing. We segment trajectories using barometric-altitude trend changes and maintain separate vector memories for each phase. For each segment, we generate a textual representation, compute embeddings using a pretrained text embedding model (text-embedding-3-large), and index them in a FAISS-based database. Phase-aware indexing prevents the mixing of incompatible motion patterns and stabilizes retrieval during inference.

D. Retrieval-Conditioned Iterative Reconstruction

Reconstruction begins from the last verified state x_t . The current flight phase is inferred from the barometric altitude, which remains trusted under GNSS interference. A structured textual summary of the verified prefix is embedded and used to retrieve the top- k phase-consistent historical trajectory segments: $C_k = \{C_1, \dots, C_k\}$.

Embeddings are L2-normalized, and retrieval uses cosine similarity, implemented as inner-product search in FAISS within the corresponding phase-specific index.

Conditioned on (i) the verified trajectory prefix, (ii) the retrieved reference segments, and (iii) the aligned meteorological context, an LLM proposes the latitude and longitude of the next aircraft state $\hat{x}_{t+1}$. The reconstructed trajectory is then updated iteratively:

$$X_{0:t+1} \leftarrow X_{0:t} \cup \{\hat{x}_{t+1}\}.$$

After each accepted state, the query summary is updated with the newly reconstructed prefix, so retrieval is repeated at every step rather than fixed from the initial prefix. This allows the retrieved neighbors to evolve as the reconstructed trajectory progresses.

In each iteration, the LLM predicts only the latitude and longitude for the next time step. No post-detection ADS-B latitude, longitude, ground-speed, or heading measurement is used as an observation. Barometric altitude is retained because it is pressure-derived rather than GNSS-derived, while ground speed and heading are inferred from the accepted reconstructed motion rather than generated by the LLM. The accepted reconstructed position is then used to query the meteorological context for the following reconstruction step. The retrieval-generation loop iterates to predict $\hat{x}_{t+2}, \dots, \hat{x}_n$ until trusted ADS-B reporting resumes or the flight concludes.

E. Phase-Specific Kinematic Validation

The admissible motion envelope is defined as a circle centered at the last accepted position with radius

$$d_{\max} = \alpha v_t \Delta t, \quad (1)$$

where v_t is the ground speed associated with the previous accepted state, using the last trusted ground speed for the first reconstruction step and the inferred reconstructed speed thereafter, $\Delta t = 120$ s is the ADS-B sampling interval, and α is a tolerance factor accommodating acceleration and measurement variability. If the LLM-proposed position lies outside this envelope, it is projected to the closest point on the boundary along the bearing from the previous accepted state, ensuring that no physically infeasible displacement can be appended to the reconstructed trajectory.

a) Calibration of α . We set $\alpha = 1.2$ based on the empirical distribution of step-to-step displacement ratios $d_{t+1}/(v_t \Delta t)$ computed across the training set. The 99th percentile of this distribution is 1.18, making $\alpha = 1.2$ a principled data-driven threshold: it is tight enough to suppress large-displacement LLM hallucinations while permitting the natural acceleration and deceleration events observed in real commercial flight. Values below $\alpha = 1.1$ over-constrain the envelope, causing physically plausible proposals to be systematically projected to the boundary and introducing a straight-line bias in the reconstructed path. Values above $\alpha = 1.4$ permit displacements that are inconsistent with normal commercial flight dynamics, reducing the validator’s ability to correct LLM drift over longer blackout intervals.

b) Adapting to different sampling intervals. Because $d_{\max}$ scales linearly with Δt , the envelope automatically tightens for higher-frequency feeds without requiring re-calibration of α . However, at shorter sampling intervals, the per-step displacement ratio distribution narrows, as aircraft have less time to accelerate between observations. For feeds with $\Delta t \leq 30$ s, we therefore recommend reducing α to the range $[1.05, 1.10]$, recalibrated from the 99th percentile of the displacement ratio distribution on a representative set of flights at the target sampling rate.

IV. EVALUATION SETUP

A. Data Collection and Preprocessing

We collected 931 commercial flight trajectories for the Paris→Rome corridor (46,550 ADS-B messages) from April 19, 2025 to June 1, 2025 using the FlightRadar24 API. Each trajectory corresponds to a single flight and consists of a time-ordered sequence of ADS-B states, where each state represents the aircraft’s reported position at a specific timestamp. Each state includes latitude, longitude, barometric altitude, ground speed, and heading (subject to availability). The trajectories contain approximately 50 states per flight, sampled at roughly 2-minute intervals along the flight path.

Meteorological context is obtained from the Meteomatics API using the current trajectory position, timestamp, and barometric altitude. During reconstruction, the current reconstructed position is used for subsequent weather queries rather than the unavailable post-detection ground-truth position. The retrieved variables include turbulence indicators and wind-related features relevant to aircraft motion. Standard preprocessing removes duplicate messages, enforces monotonically increasing timestamps, and interpolates short temporal gaps.

Each trajectory is then segmented into takeoff, cruising, and landing phases, and all evaluation results are reported separately for each phase.

We use a chronological split to avoid temporal leakage: 831 trajectories (April 19 to May 24) for training/validation and 100 trajectories (May 25 to June 1) for held-out testing.

B. Spoofing Corruption Protocol

We evaluate TrajRAG under a post-detection recovery framework. While GNSS interference can manifest in various forms—including gradual positional drift, velocity manipulation, or “lock-on” spoofing—our protocol adopts a safety-aligned, attack-agnostic stance. Once spoofing is detected at time t , we trigger a data-blackout protocol for all subsequent horizontal position-related fields in $X_{t+1:n}$. By discarding these observations rather than filtering corrupted signals, reconstruction remains grounded in trusted historical patterns and the verified pre-spoofing state. This approach ensures robustness against unseen or sophisticated spoofing signatures that might bypass traditional smoothing filters. To evaluate this, we apply simulated corruption to real commercial flight trajectories by randomized spoofing onset times and corruption durations, and validate the reconstructed trajectories against the original uncorrupted trajectories as the ground truth. Corruption durations range between 5 and 15 time steps long (approximately 10–30 minutes at a 2-minute sampling rate), covering both short and extended blackout scenarios.

C. Baselines

We compare against five baseline groups. First, model-based and statistical baselines include Kalman filtering and ARIMA. Second, recurrent neural sequence baselines include LSTM and BiLSTM trained on the same chronological training split. Third, modern time-series forecasting baselines include DLinear [23], PatchTST [24], and iTransformer [25]. We include iTransformer in particular because inverted-Transformer architectures have recently shown strong performance for multivariate time-series forecasting and have been adopted for aircraft trajectory prediction. To ensure fairness, all neural and time-series forecasting baselines receive the same phase-segmented windows used by TrajRAG, and phase is included as an input feature where applicable. Fourth, retrieval-only baselines predict the next state by averaging the corresponding future states of the top-k retrieved historical trajectories. Fifth, we include a numerical retrieval baseline, Numeric-kNN, which retrieves neighbors using standardized prefix vectors rather than textual embeddings. Specifically, Numeric-kNN represents each prefix using latitude, longitude, barometric altitude, speed, heading, phase, and weather variables, and then applies the same top-k future-state averaging used by the retrieval-only baseline. This directly tests whether TrajRAG’s textual state abstraction improves over conventional numeric similarity.

D. Metric

We report reconstruction accuracy using the *mean Haversine distance error* (MHE) [32], which measures the average

great-circle distance between reconstructed and ground-truth positions over the corrupted interval:

$$\text{MHE} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} d_{\text{hav}}(\hat{\mathbf{x}}_t, \mathbf{x}_t), \quad (2)$$

where $\mathcal{T}$ are spoofed indices, $\hat{\mathbf{x}}_t$ is the reconstruction, and $\mathbf{x}_t$ is the ground truth.

E. Mixed-Route Memory Robustness

We evaluate mixed-route memory robustness using an additional 613 Barcelona→Munich trajectories collected from December 17, 2025, to January 18, 2026, under two memory settings:

(S1) In-route memory. The vector memory is constructed from historical flights for the same route as the test trajectory (e.g., Barcelona→Munich train, Barcelona→Munich test).

(S2) Mixed-route memory (deployment setting). The vector memory is constructed from the union of training trajectories across all routes, while testing is performed per route. This setting reflects a realistic deployment scenario in which the framework stores historical flights from multiple corridors over time in the same index.

F. Ablation Studies

To validate the design principles underlying TrajRAG, we conduct a series of targeted ablation studies. Each one isolates a core component of the pipeline and empirically evaluates its contribution to reconstruction accuracy and overall framework stability under GNSS interference. We ablate key components of TrajRAG by removing retrieval, disabling iterative feedback, removing weather context, removing the kinematic validation, removing the LLM generation, and replacing the textual representation with numerical embeddings. We additionally evaluate sensitivity to the LLM backbone while keeping retrieval, prompts, and validation fixed. For ablation comparisons, statistical significance is computed using paired two-sided Wilcoxon signed-rank tests over per-flight reconstruction errors, comparing each ablated configuration against the full TrajRAG pipeline on the same test flights. We report uncorrected p-values because the tests are used as diagnostic evidence for component contribution rather than as independent hypothesis claims.

G. Implementation Details

Unless otherwise specified, all main results use gpt-5.1 as the generation backbone, with text-embedding-3-large for embeddings, $K = 5$ retrieved neighbors, and a FAISS index per flight phase. We select gpt-5.1 as the default backbone because it provides the strongest overall accuracy–cost trade-off in our sensitivity analysis (Section V-F): it achieves the lowest error on the dynamic takeoff and landing phases, while being the cheapest of the frontier models evaluated. The effect of varying this choice is reported in Table III.

V. RESULTS

A. Phase-Wise Error Distribution

Trajectory reconstruction difficulty is phase-dependent. TrajRAG obtains the largest errors during takeoff, where short segments and rapid dynamics provide limited context, and the smallest errors during landing. This motivates reporting all results by phase rather than only as a single aggregate score.

B. Comparison to Baselines

Table I reports the MHE for each reconstruction method across flight phases. TrajRAG consistently achieves the lowest error across all phases. Classical approaches such as Kalman filtering and ARIMA degrade substantially under data blackout conditions due to their reliance on continuous observations. Recurrent neural baselines (LSTM and BiLSTM) improve over these classical methods but remain less accurate than TrajRAG and exhibit larger deviations, especially in takeoff and cruising.

The modern time-series forecasting baselines provide a stronger and more recent comparison. DLinear, PatchTST, and iTransformer reduce error in some phases relative to classical and recurrent models, with iTransformer being the strongest Transformer-style baseline in cruising. However, all three remain substantially less accurate than TrajRAG under the post-detection blackout protocol. This gap is most visible in landing, where procedural regularity allows retrieval-grounded reconstruction to remain highly accurate, while pure forecasting baselines still accumulate open-loop error. These results show that even strong modern forecasting architectures, including iTransformer, are not sufficient by themselves when corrupted horizontal observations must be discarded rather than continuously assimilated.

The retrieval baselines isolate the source of TrajRAG’s advantage: mean-retrieval and Numeric-kNN both improve over several forecasting baselines, confirming that historical grounding is important. However, both remain less accurate than TrajRAG, indicating that retrieval-conditioned generation contributes additional gains beyond neighbor averaging alone. Overall, the results indicate that retrieval-grounded reconstruction provides a more stable mechanism than either pure forecasting or retrieval-only averaging when direct trajectory observations are unavailable.

C. Retrieval Size K Selection

The number of retrieved neighbors K affects both reconstruction accuracy and computational cost. We analyze MHE as a function of $K \in \{1, \dots, 20\}$. Increasing K initially improves performance, as aggregating multiple retrieved trajectories reduces sensitivity to individual retrieval errors. However, when K becomes too large, reconstruction quality deteriorates because fewer structurally similar trajectories are included in the retrieval set. We therefore set $K = 5$ in all experiments, which provides a stable balance between accuracy and efficiency.

TABLE I
COMPARATIVE PERFORMANCE ACROSS FLIGHT PHASES. HAVERSINE DISTANCE ERROR MEAN AND STANDARD DEVIATION ARE REPORTED IN KM.

Method	Takeoff		Cruising		Landing	
	μ	σ	μ	σ	μ	σ
Kalman	60.95	30.45	39.06	25.10	112.61	31.78
ARIMA	112.34	14.00	37.69	37.57	73.82	22.55
LSTM	33.78	23.11	40.66	22.72	18.62	14.33
BiLSTM	35.79	31.58	46.83	27.71	11.17	4.91
Mean-Retrieval	48.94	37.77	30.80	21.64	8.52	6.42
Numeric-kNN	66.19	29.34	32.49	17.12	16.92	10.20
DLinear	43.99	13.18	21.01	7.03	35.47	11.06
PatchTST	55.77	10.80	23.04	8.90	38.17	9.28
iTransformer	47.89	8.08	19.79	5.40	37.58	7.35
TrajRAG	20.78	4.60	9.84	4.40	2.46	2.33

TABLE II
PER-STATE RECONSTRUCTION LATENCY AND ESTIMATED GENERATION COST.

Phase	Text (s)	Ret. (s)	Gen. (s)	Total (s)	Cost (\$)
Takeoff	3.2	0.53	1.5	5.23	.014
Cruise	4.0	0.51	2.3	6.81	.018
Landing	3.4	0.49	1.7	5.59	.016

D. Latency Analysis

Table II reports the latency (s) and estimated cost (\$) required to reconstruct a single missing ADS-B state (i.e., one timestamp in the trajectory) for each flight phase. The text-construction step is incurred once at blackout onset; the repeated iterative step consists mainly of retrieval and generation and remains below 3 seconds per ADS-B state.

Excluding the one-time textual representation construction, the per-state iterative latency is approximately 2.0–2.8 seconds across the flight phases. Given that ADS-B messages in our dataset are sampled at approximately 120-second intervals, this computational overhead remains well within real-time operational constraints. The cost values are estimated using the LLM’s API pricing and correspond to the token usage required to generate a single reconstructed state. In our setup, the per-state generation cost remains below \$0.02, supporting scalable deployment in continuous surveillance settings.

E. Data Efficiency

We evaluate data efficiency by comparing the MHE of TrajRAG against the recurrent LSTM baseline and the strongest modern forecaster, iTransformer, as the reference set size grows from roughly 50 to 800 trajectories (Fig. 2). Across all three phases, TrajRAG achieves the lowest error and stabilizes earliest, with performance largely plateauing after approximately 400–500 flights. iTransformer is consistently the strongest parametric baseline and improves smoothly as more training data are added, but remains above TrajRAG in all phases. LSTM exhibits higher variance and remains less accurate even with the full training set. These results

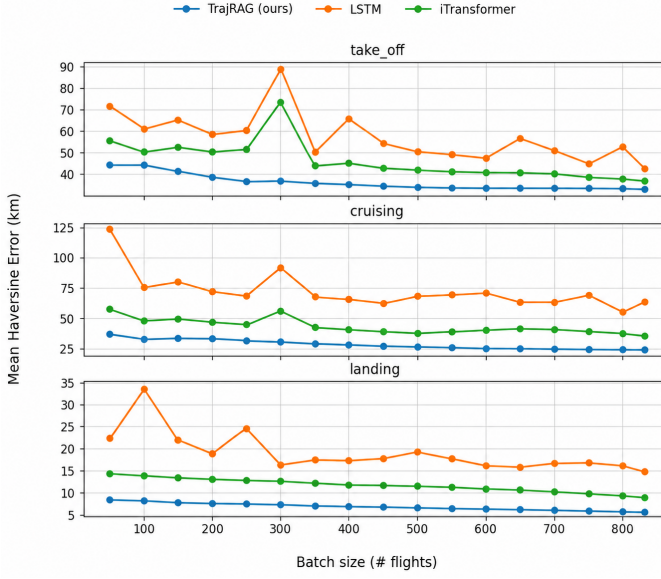


Fig. 2. Data-efficiency comparison under reduced-memory training settings. Each point reports MHE when the available reference/training set is restricted to the indicated number of flights.

TABLE III

COMPARATIVE PERFORMANCE OF LLMs ACROSS FLIGHT PHASES. MEAN AND STANDARD DEVIATION ARE REPORTED IN KM. COST IS REPORTED IN USD PER 1M INPUT TOKENS AT THE TIME OF EXPERIMENTATION.

Model	Takeoff		Cruising		Landing		Cost
	μ	σ	μ	σ	μ	σ	
gpt-4.1	25.14	5.01	7.13	3.46	2.98	2.14	2.00
gpt-4o	22.47	5.06	10.88	7.55	13.85	12.41	2.50
gpt-4o-mini	86.80	39.10	47.70	23.99	19.65	15.47	0.15
gpt-5.1	20.78	4.60	9.84	4.40	2.46	2.33	1.25
gpt-5.2	20.82	4.86	7.12	3.32	2.62	2.53	1.75
Llama 3.3 7B	52.89	13.22	52.16	13.04	13.43	3.36	0.27
Mistral 7B	59.79	14.95	58.96	14.74	15.18	3.80	0.10
Qwen3-Max-Think.	41.39	10.35	40.82	10.21	10.51	2.63	0.20

support the benefit of inference-time grounding through phase-aware retrieval, particularly when route coverage is limited and retraining data are scarce.

F. Sensitivity to the Underlying LLM

We evaluate the impact of the underlying LLM by executing the TrajRAG pipeline with several models while keeping retrieval, prompts, and all other components fixed. The results show that stronger reasoning models produce more stable reconstruction behavior, while smaller models exhibit higher variance. Table III reports the corresponding performance and cost comparison.

G. Mixed-Route Memory Robustness

Table IV evaluates whether expanding the retrieval memory with trajectories from additional routes degrades reconstruction accuracy. S1 denotes a route-specific memory containing only trajectories from the evaluated route, while S2 denotes a mixed-route memory containing trajectories from multiple

TABLE IV

EFFECT OF MIXED-ROUTE MEMORY: MHE CHANGE FROM S1 TO S2 (KM).

Route	Takeoff	Cruise	Landing
Paris–Rome	+0.09	-0.11	+0.71
Barcelona–Munich	+0.74	-0.02	-0.06

corridors. We report the change in MHE from S1 to S2 for each route and flight phase. Across all route-phase combinations, replacing route-specific memory with mixed-route memory changes MHE by less than 1 km. This indicates that retrieval remains phase-consistent and route-relevant even when the vector index contains trajectories from multiple corridors, supporting scalable deployment across multi-route operational settings.

H. Ablation Studies

To quantify the contribution of each component, we evaluate TrajRAG with one component removed at a time on the same test set used in the main evaluation.

Table V shows that retrieval is the main driver of performance. Removing retrieval causes the largest degradation in every phase, increasing the mean error from 20.78 km to 68.70 km in takeoff, from 9.84 km to 38.04 km in cruising, and from 2.46 km to 36.93 km in landing. This confirms that historical grounding is essential when GNSS-derived horizontal positions are unavailable.

The LLM proposal step also adds value beyond direct neighbor averaging. Replacing the LLM with mean retrieval significantly worsens performance in all phases, especially takeoff and cruising, showing that retrieval-conditioned generation adapts the retrieved evidence better than uniform top- k averaging. The feedback loop is similarly important: disabling it increases error and variance, confirming that updating the query after each reconstructed state helps reduce drift over the blackout interval.

The remaining components provide more targeted benefits. Textual representation significantly improves cruising and landing, where longer segments allow richer semantic matching. The kinematic validator has limited effect on mean error, but reduces variance and worst-case deviations, acting mainly as a safety layer against rare infeasible proposals. Weather and barometric-altitude context provide smaller, phase-dependent gains, mainly improving stability in more difficult segments. Overall, the ablations support the full TrajRAG design: retrieval provides the core grounding signal, the LLM improves over simple averaging, feedback reduces drift, and validation limits failure cases.

VI. REAL-WORLD MULTI-REGION CASE STUDIES

The quantitative evaluation in Section V relies on simulated corruption because computing reconstruction error requires a trusted reference trajectory. During genuine GNSS spoofing, the aircraft’s broadcast horizontal position may itself be corrupted, leaving no per-state ground truth. To complement

TABLE V

ABLATION STUDIES OF TRAJRAG COMPONENTS ACROSS FLIGHT PHASES. THE P-VALUE IS RELATIVE TO THE FULL PIPELINE. “W/O LLM (MEAN RETRIEVAL)” REPLACES THE LLM PROPOSAL WITH UNIFORM TOP- k NEIGHBOR AVERAGING WHILE RETAINING THE ITERATIVE LOOP AND KINEMATIC VALIDATOR; IT IS THEREFORE MEAN-RETRIEVAL EVALUATED UNDER THE MATCHED PIPELINE, AND THE GAP TO THE FULL PIPELINE ISOLATES THE LLM CONTRIBUTION.

Phase	Configuration	Mean (km)	Std.	Max (km)	p-value
Takeoff	w/o retrieval	68.70	26.69	125.1	3.0×10^{-8}
	w/o LLM (mean retrieval)	48.94	37.77	88.45	6.9×10^{-6}
	w/o kinematic validation	22.03	6.90	35.64	0.764
	w/o feedback loop	46.57	32.81	187.7	6.2×10^{-6}
	w/o textual representation	35.56	40.71	219.1	0.078
	w/o weather data	23.47	17.55	97.1	0.810
	w/o barometric altitude	20.56	7.28	33.6	0.755
	TrajRAG (ours)	20.78	4.60	28.8	–
Cruising	w/o retrieval	38.04	20.64	83.7	5.7×10^{-7}
	w/o LLM (mean retrieval)	30.80	21.64	59.39	0.002
	w/o kinematic validation	9.99	9.96	20.36	0.356
	w/o feedback loop	34.76	14.75	65.3	3.0×10^{-8}
	w/o textual representation	29.31	22.70	74.1	0.001
	w/o weather data	13.81	9.87	37.5	0.307
	w/o barometric altitude	20.51	16.57	63.9	0.015
	TrajRAG (ours)	9.84	4.40	14.8	–
Landing	w/o retrieval	36.93	27.62	129.3	1.5×10^{-7}
	w/o LLM (mean retrieval)	8.52	6.42	15.51	0.008
	w/o kinematic validation	2.46	2.45	4.78	0.564
	w/o feedback loop	5.87	3.74	14.5	0.001
	w/o textual representation	4.67	3.27	12.7	0.017
	w/o weather data	3.53	2.10	10.1	0.355
	w/o barometric altitude	4.90	3.78	13.9	0.014
	TrajRAG (ours)	2.46	2.33	4.6	–

the controlled benchmark, we evaluate TrajRAG on four real-world MLAT-supported interference episodes across distinct geographic regions.

A. Flight Selection and Setup

We screened operational flights in the Flightradar24 (FR24) feed whose data-source label switched to `Terrestrial MLAT`, indicating that the displayed positions were derived from an independent Time Difference of Arrival (TDOA) solution across FR24’s ground-receiver network rather than from the aircraft’s GNSS broadcast [33], [34]. Candidates were retained only when a valid barometric-altitude trace was available throughout the interference window. ADS-B messages are sampled at approximately 2-minute intervals, consistent with the controlled benchmark.

We selected four MLAT-supported events covering the Gulf/Saudi Arabian, Black Sea/Northern Turkey, Baltic, and Sinai–Red Sea/Gulf of Aqaba corridors, spanning passenger and cargo flights, two aircraft families, and windows of 46–103 minutes. Table VI summarizes the cases.

For each flight, the retrieval memory was populated with 200 historical trajectories from the same route corridor, collected during May 2025 and applied at reconstruction time in May 2026. Trajectories were screened to exclude prolonged surveillance degradation, GPS-altitude loss, or spoofing artifacts, and the affected flight and event date were never included in memory.

B. Reconstruction Results

Reconstruction begins at the first point where the FR24 source label switches to `Terrestrial MLAT`. TrajRAG receives the verified pre-interference prefix, trusted barometric altitude, route-specific memory, and aligned meteorological context, then iteratively reconstructs horizontal positions at 2-minute intervals under the same kinematic constraints used in the controlled benchmark. Only fixes explicitly labeled `Terrestrial MLAT` are retained as the independent consistency reference; fixes labeled `ADS-B` inside the interference window are discarded.

Figure 3 shows the detailed AMM–AUH case. The corrupted ADS-B broadcast exhibits abrupt lateral displacement and stationary clustering inconsistent with the aircraft’s prior heading and speed, while TrajRAG reconstructs a kinematically coherent path that closely tracks the retained MLAT reference positions. Table VII reports consistency metrics across all four cases. Mean MLAT deviation remains below 8 km across all events, and endpoint deviation stays below 6.4 km even after the longest 103-minute blackout.

C. Limitations

These real-world episodes should not be interpreted as a full population-level accuracy estimate. MLAT accuracy depends on ground-receiver geometry, synchronization quality, and regional receiver density and therefore does not provide the same type of per-state ground truth as the controlled benchmark.

TABLE VI
REAL-WORLD MLAT-SUPPORTED GNSS-INTERFERENCE CASES.

Flight	Route	Aircraft	Region / onset location	Window	Memory
A6-AQF	AMM-AUH	A320	Saudi Arabian interior, (28.22°N, 41.82°E)	103 min	200
TG941	MXP-BKK	B789	Black Sea / N. Turkey, (42.19°N, 35.15°E)	46 min	200
CK602	BUD-CKG	B77L	Baltic region, (56.97°N, 22.61°E)	71 min	200
QR8213	DOH-MAD	B77L	Sinai-Red Sea / Gulf of Aqaba, (29.07°N, 34.69°E)	53 min	200

TABLE VII
REAL-WORLD RECONSTRUCTION CONSISTENCY CHECKS (ALL DISTANCES IN KM).

Flight	Route	Window	Mean MLAT dev.	Std.	Max MLAT dev.	Endpoint dev.
A6-AQF	AMM-AUH	103 min	6.96	2.57	9.70	1.97
TG941	MXP-BKK	46 min	7.63	3.94	13.60	4.78
CK602	BUD-CKG	71 min	7.95	4.27	12.98	5.01
QR8213	DOH-MAD	53 min	6.95	4.69	11.29	6.36

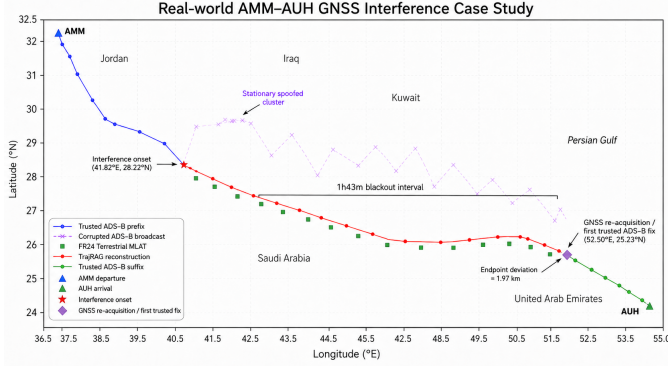


Fig. 3. AMM-AUH flight under GNSS Interference over the Saudi Arabian interior. The trusted ADS-B prefix (blue) ends at the interference onset. The corrupted ADS-B broadcast (purple dashed) shows physically implausible behavior. TrajRAG (red) reconstructs a coherent path consistent with the independent FR24 Terrestrial MLAT reference (green squares).

In addition, the four selected cases cover diverse regions and aircraft types but remain a small operational sample. We therefore use MLAT as an independent consistency reference: the goal is to test whether TrajRAG produces trajectories that remain coherent with a non-GNSS positioning source during authentic interference episodes. Under this interpretation, the consistent sub-8 km mean MLAT deviation and sub-7 km endpoint deviation provide practical corroboration that the controlled-benchmark behavior transfers to real GNSS-interference settings.

VII. DISCUSSION

Applied lesson 1: phase constraints matter. The results show that reconstruction difficulty is phase-dependent: takeoff is short and dynamic, cruise is stable but long-horizon, and landing benefits from stronger procedural regularity. This supports the use of phase-specific memory rather than a single global index.

Applied lesson 2: retrieval must be grounded and validated. The ablations show that retrieval is the dominant contributor, while the feedback loop and kinematic validator reduce drift and worst-case failures. In an aviation setting, the

LLM should therefore be treated as a proposal generator rather than an authority.

Operational integration. TrajRAG is intended as a post-detection mitigation module. When a GNSS interference detector flags an aircraft, corrupted horizontal-position fields are discarded, reconstruction starts from the last verified state, and generated states are provided with a low-confidence flag when they cannot be reconciled with feasible motion or resumed trusted observations.

VIII. CONCLUSION

We presented TrajRAG, a retrieval-augmented framework for aircraft trajectory reconstruction during GNSS interference intervals. By grounding reconstruction in historically observed, phase-consistent flight behavior, the framework reframes trajectory recovery as a retrieval-grounded sequential inference problem rather than pure extrapolation.

Across multiple flight phases and air corridors, TrajRAG demonstrates consistent improvements in reconstruction accuracy and, critically, substantial reductions in worst-case deviations, compared to classical filtering, statistical forecasting, recurrent sequence models, modern time-series forecasting baselines including DLinear, PatchTST, and iTransformer, and retrieval-only alternatives. The results highlight the importance of phase-aware modeling, structured semantic representation, and retrieval grounding to stabilize long-horizon prediction under data blackout conditions.

Unlike fully parametric predictors, TrajRAG adapts and scales to new data through memory expansion rather than retraining, enabling incremental integration of new routes and evolving traffic patterns.

IX. REPRODUCIBILITY

To support transparency and independent verification, we release the complete experimental suite at <https://github.com/yf2595/TrajRAG>. The repository contains the full TrajRAG pipeline (offline memory construction, retrieval-conditioned reconstruction, and kinematic validation), all baseline implementations (Kalman, ARIMA, LSTM, BiLSTM, DLinear, PatchTST, iTransformer, Mean-Retrieval, and Numeric-kNN), the spoofing-corruption protocol, evaluation scripts, and the

configuration files needed to regenerate every table and figure in this paper. Model identifiers, embedding settings, retrieval parameters ($K = 5$), and kinematic-validator parameters ($\alpha = 1.2$, $\Delta t = 120$ s) are specified in versioned configuration files, ensuring that each reported result maps to a specific run configuration. The released material includes the data samples (ADS-B trajectories and aligned weather context), the chronological train/test splits, and the randomized corruption settings (onset times and blackout durations) used to produce the reported results. Where full datasets cannot be redistributed due to provider terms, we document the exact API queries against the FlightRadar24 and Meteomatics services required to reconstruct the corpus, including the corridors, date ranges, and per-state query parameters (current trusted or reconstructed position, timestamp, and barometric altitude). Reproducing the LLM-backed experiments requires an OpenAI API key for the GPT-family backbones (e.g., gpt-5.1, used for the main results) and for the text-embedding-3-large embedding model. Reproducing the open-weight backbones reported in our LLM sensitivity analysis (e.g., Llama 3.3, Mistral 7B, Qwen3-Max) requires downloading the corresponding model weights.

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